



HEPZIBHA SUNDARARAJ

Sr. Front-End | API | Full Stack Developer | React + Python | Java | Node JS | Generative AI | AI Agents | Agentic AI | MVC Server | RAG | AWS Bedrock | UI-UX Designer |
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PROFESSIONAL SUMMARY

Results-driven Full Stack Developer with 15+ years of experience with 5+ years of technical Architect, leadership and building scalable enterprise SaaS applications using React, Next JS, TypeScript, Vue.js, Tailwind CSS, Python, Java, Spring Boot, TypeScript, and Node.js. Proven track record architecting agentic AI workflows integrating OpenAI, Lang Chain, Claude Code, and Codex into production-grade web platforms. Strong experience in AI/ML application development, ML/NLP integration into frontend, NLP & Document Intelligence, and MLOps collaboration. Implemented type-safe, nested routing with TanStack Router across a 50+ page application, eliminating runtime routing errors through compile-time route validation and search-param type inference. Deep expertise in micro-frontend architecture, RESTful/GraphQL API design, cloud-native deployments on AWS with Kubernetes and Docker, and leading cross-functional teams through Agile delivery at Fortune 500 companies including Freddie Mac, State Street Bank, and Johnson & Johnson. Engineering async/latency-aware UX patterns, including loading states, retry logic and partial-response handling, to ensure smooth user experience during variable AI response times. Developed AI/ML and GenAI powered applications using Chainlit and Streamlit, implementing complex state management and real-time data updates for interactive user experience.

CORE COMPETENCIES

Frontend: React JS, Next JS, TypeScript, JavaScript (ES6+), Material UI, Redux, Vue.js, HTML5, CSS3, SASS
Backend: Java, Spring Boot, Microservices, Node.js, Express, Python Flask, Fast API, Django, REST API, GraphQL
Agentic AI & LLM: OpenAI API, LLM, Lang Graph, Lang Chain, Claude Code, Codex, Cursor, GitHub Copilot, NLP, RAG, tool/function calling, and agentic workflows
Cloud & DevOps: AWS (Lambda, EC2, S3, ECS, Code Pipeline, CloudWatch, Bedrock), SAP cloud, Kubernetes, Docker, Jenkins, GitLab CI/CD, Maven
Databases: SQL Server, MySQL, MongoDB, Oracle, Snowflake (Materialized Views, Snowpark, Data Sharing), SAP HANA
Data & Analytics: Power BI, Tableau, SAP BO, D3.js
Testing: Jest, Cypress, Playwright, React Testing Library, Py Test, WCAG 2.1, Section 508
UI/UX & CMS: Figma, Adobe XD, Sketch, Storybook, SharePoint CMS

PROFESSIONAL EXPERIENCE

Sr. React / FullStack Developer / React + AI Lead | Freddie Mac / Conduent (Birla Soft) | Feb 2022 – Present

- Architected and deployed scalable RESTful APIs using Node.js and Express, handling 10K+ concurrent requests with sub-100ms response times through async/await patterns and connection pooling.
- Designed and developed UI/UX for analytics dashboards and interactive reporting tools, translating complex data sets into intuitive, actionable visualizations for end users.
- Architected and implemented a Flex-based unidirectional data flow to manage complex application state, reducing debugging time and improving predictability across 15+ interconnected components,
- Architected and shipped a full-stack web application using TanStack start, leveraging its file-based routing and SSR capabilities to reduce initial page load time while maintaining full type safety across client and server.



- Built production grade React / TypeScript components and admin surfaces – functional components, hooks, Redux Toolkit state management for an enterprise healthcare CLAIM portal, including sortable, filterable data tables, multi-step modal wizards, and dashboard UI
- Designed and developed intuitive front-end interfaces for AI-powered products including chatbots, copilots, and RAG-Retrieval Augmented Generation applications, translating complex model outputs into seamless user experiences.
- Built a GenAI chat application using Streamlit and Chainlit, implementing streaming token responses, session-based state management, and real-time UI updates for a conversational analytics interface.
- Designed a Retrieval-Augmented Generation(RAG) pipeline with LangChain and OpenAI, grounding LLM responses in enterprise-style datasets to reduce hallucination and improve answer accuracy.
- Developed a natural language to SQL module that translates plain-English questions into validated SQL queries against a Snowflake backend dataset, returning both the generated query and result set to the user.
- Built a lightweight semantic model / data dictionary layer mapping business terms like revenue, active customer to underlying table and column definitions, improving NL to SQL accuracy and consistency.
- Implemented asynchronous UX patterns loading indicators, retry logic and partial response streaming to handle latency gracefully during LLM and query execution.
- Added explainable AI elements including confidence scoring and source query transparency, so users could see why an answer was generated.
- Deployed the application using Docker and AWS, with CI/CD automation via GitHub Actions for testing and container builds.
- Configured and optimized webpack build pipeline, reducing production bundle size through code splitting, tree shaking and lazy loading.
- Migrated frontend build tooling from Create React App to Vite, upgraded React, TypeScript, and React Router to current major versions, improving bundle optimization and environment variable management.
- Built and maintained responsive, component-based UI features using Vue.js and Vuex, applying reusable component architecture and state management patterns directly transferable to React's component/hooks based ecosystem, improving development velocity across the product modules.
- Migrated a legacy React application from React Router and custom fetch hooks to TanStack Router and TanStack Query, cutting boilerplate code and improving developer velocity on new feature delivery,.
- Reduced redundant API calls and integrating TanStack Query for server state management, implementing intelligent caching, background refetching, and optimistic UI updates across critical user flows.
- Built a real time dashboard using TanStack Start's server functions and TanStack Query's mutation and invalidation pipelines, enabling seamless data synchronization between client and server with automatic catch updates.
- Implemented OAuth2, OLDC authentication and SCIM, IDM integrations using ForgeRock Access Management – AM and identity Management -IDM, maintaining journey flows for login, MFA, email, phone validation, password update, and identity verification with role-based access control.
- Adopted a new design system with design tokens, CSS custom properties, Grid, Flexbox layouts, and utility classes, removing legacy CSS-in-JS while ensuring WCAG 2.1 AA accessibility- ARIA, keyboard navigation from Figma prototypes.
- Deployed and validated containerized frontend workloads- Docker, Nginx, SPA Service config across Kubernetes namespaces on AWS EKS in dev, sandbox, and UAT environments, coordinating CI/CD pipeline integration with DevOps
- Integrated LLM powered features into React applications by consuming APIs from Open AI, Azure OpenAI, and Hugging Face, enabling real-time AI-driven integrations for end users.
- Built and optimized streaming response interfaces for conversational AI assistants, improving perceived latency and user engagement during real time LLM output generation.
- Collaborated with AI/ML engineers to visualize model predictions, embeddings, and confidence scores in React dashboards, bridging the gap between backend ML pipelines and user-facing analytics.



- Applied prompt engineering principles when designing UI workflows for AI assistants, ensuring consistent, high quality model interactions across the application.
- Developed front-end components for NLP driven features such as intelligent search, document processing, and text classification, improving information retrieval accuracy for end users.
- Built recommendation system interfaces powered by ML models, increasing user engagement through personalized, data-driven content suggestions.
- Collaborated with MLOps teams to build front-end monitoring dashboards for model performance metrics – accuracy, latency, drift, enabling faster identification of degraded AI outputs in production applications.
- Integrated CI/CD pipelines connecting front-end deployments with ML model versioning and rollout schedules, ensuring seamless delivery of updated AI/ML feature without disrupting user-facing applications.
- Led a team of frontend engineers across the full SDLC of enterprise-scale web applications, architecting and delivering React-TypeScript-Next.js solutions that improved performance, scalability, and maintainability while mentoring developers and driving technical decisions.
- Built and maintained robust CI/CD pipelines on Amazon DevOps, integrating automated testing with Playwright and Cypress to reduce release cycle time and catch regressions before production, improving overall deployment reliability.
- Leveraged Next.js for server-side rendering and improved performance of AI powered web applications.
- Integrated vector database query results into React front-end for semantic search and RAG-based applications.
- Gained working familiarity with Lang Chain- Llama Index workflows to better collaborate with AI engineering teams on RAG application architecture.
- Optimized Node.js application performance by implementing worker threads, clustering, and caching strategies, reducing CPU load time and cutting average API response time.
- Drove adoption of AI-assisted development workflows using GitHub Copilot, Cursor, and Claude to accelerate code generation, refactoring, and code review, boosting team productivity and reducing time-to-delivery on key features.
- Owned technical leadership responsibilities including architecture decisions, code reviews, and sprint and sprint planning for React-based enterprise applications, partnering with product and QA teams to ensure high-quality, scalable frontend delivery.
- Wrote comprehensive unit and shallow rendering tests using Enzyme, achieving 95% test code coverage across core UI components.
- Refactored legacy Enzyme test suites to improve reliability and reduce flaky test failures in CI/CD pipelines.
- Led adoption of AI-driven development practices using Cursor and GitHub Copilot within a Node.js codebase, accelerating feature delivery by 35% and reducing boilerplate authoring time by 70%.
- Built a full-stack AI Agent interface using Next.js 16 App Router and React 19 integrating @ag-ui/client to enable real-time bidirectional communication between frontend and Lang Graph-orchestrated agents.
- Designed and implemented agentic AI workflows integrating OpenAI and Lang Chain APIs into enterprise SaaS platforms, enabling intelligent text generation, context-aware chat, and AI-assisted code features for 50,000+ users — leveraging Claude Code and Codex as primary AI coding agents throughout development.
- Built AI-first full-stack features using React, TypeScript, and Java Spring Boot microservices, integrating LLM tool/function calling and RAG pipelines to automate complex enterprise SaaS workflows safely and at scale.
- Developed Python Fast API backend with Lang Chain / Lang Graph for multi-step AI orchestration, implementing A2A SDK agent communication and token-based authentication with OpenAI/Horizon language models.
- Engineered a responsive, type-safe UI in TypeScript with Tailwind CSS 4.x and Daisy UI components, including markdown rendering via React Markdown with remark-gfm for rich agent response display.



- Architected Lang Graph agent workflow to coordinate multi-agent pipelines, handling state management, conditional routing, and tool-use chains to power complex AI-driven user interactions.
- Owned end-to-end TypeScript/Python type safety across a distributed AI system, aligning Fast API pydantic schemas with frontend TypeScript interfaces to eliminate runtime errors in agent communication payloads.
- Architected and led development of micro-frontend React applications with modular, independently deployable components, defining coding standards, architectural roadmaps, and best practices while mentoring junior engineers and leading code reviews.
- Designed and built a scalable library of reusable Material UI (MUI) components including custom-themed buttons, modals, data grids, and form controls, reducing UI development time by 30% across multiple enterprise React applications.
- Familiarity with AI applications, LLM integration, agentic workflows, or strong interest in learning.
- Experience using AI coding assistants such as Codex, Claude Code, Cursor, Copilot and similar tools.
- Integrated RESTful and GraphQL APIs into MUI-based React dashboards using Axios interceptors and Redux state management, delivering real-time data fetching with error boundaries and seamless loading states.
- Built fully responsive, mobile-first web applications using MUI Grid/Box layout with custom useMediaQuery hooks, ensuring consistent cross-device experience across desktop, tablet, and mobile viewports.
- Engineered centralized MUI theme configuration using createTheme and ThemeProvider, enforcing brand-consistent typography, color palettes, and component overrides across the enterprise front-end suite.
- Optimized MUI-driven React applications via component memoization, lazy loading, and code splitting, enforcing WCAG 2.1 accessibility standards including keyboard navigation, ARIA roles, and color contrast compliance.
- Implemented explainable AI features such as confidence scores, source citations, and disclaimers, improving user trust and transparency in AI-generated outputs.
- Automated CI/CD pipelines with Jenkins, Docker, Kubernetes, and AWS Lambda/S3; enforced security via IAM, KMS, and CloudTrail achieving zero breaches across all deployments.
- Developed and integrated front-end applications with SharePoint CMS, enabling dynamic content delivery, template-driven UI components, and scalable content workflows across multiple enterprise platforms.
- Experience with AI Agents, RAG, tool/function calling, and workflow automation.
- Experience with modernizing enterprise SaaS applications, MongoDB, Jenkins, GitLab, and similar platforms.

Sr. React / Full Stack Developer | State Street Bank, Boston, MA | Aug 2019 – Jan 2022

- Modernized legacy enterprise SaaS applications by migrating Angular apps to React and Next JS, improving load time, long-term maintainability, and alignment with modern full-stack architecture standards.
- Built and optimized large-scale React SPAs and mobile apps using modern JavaScript (ES6+), Redux Saga, and advanced state management patterns, delivering pixel-perfect, ADA-compliant, responsive user interfaces.
- Architected and built scalable, highly customized production grade interfaces in React, maintaining performance and consistency across large-scale component libraries and design systems.
- Applied prompt design principles to optimize AI interactions, balancing response quality, latency, and user experience within chat-based interfaces.
- Built responsive AI chat interfaces with real-time streaming and token-level response handling, enabling fluid, low-latency conversational experiences.
- Developed AI-assisted features including autocomplete, summarization, and recommendation systems, integrating LLM outputs seamlessly into existing React component architectures.
- Collaborated cross-functionally with data scientists, ML engineers, and product teams to translate AI/ML capabilities into polished, user-facing interfaces.



- Optimized frontend performance for data-heavy dashboards and AI driven interfaces, reducing load times and improving responsiveness under real-time data streams.
- Collaborated on migrating legacy Vue.js components to a modern JavaScript and TypeScript framework, strengthening cross-framework proficiency in reactive UI patterns, virtual DOM concepts, and component lifecycle management core skills shared across React and Vue.
- Integrated Node.js and Express backends with REST and GraphQL APIs for real-time data flow; authored advanced SQL queries against Snowflake materialized views, leveraging Snowflake data sharing to provide governed, live access to multi-tenant datasets across enterprise clients.
- Embedded Power BI reports within React dashboards connected to Snowflake data pipelines, enabling real-time self-service analytics and KPI visualization for business stakeholders.
- Implemented robust testing with Jest, React Testing Library, Mocha, and Enzyme; integrated Splunk for performance monitoring and reliability tracking across enterprise deployments.
- Managed serverless deployments on AWS EC2, S3, Lambda, and API Gateway; championed accessibility-first development culture ensuring WCAG 508 compliance for government and enterprise clients.

UI Developer / React JS Developer | Johnson & Johnson, New Brunswick, NJ | May 2017 – Jul 2019

- Led front-end technical initiatives as a technical lead across micro-frontend React applications integrated with Java-based microservices and Python Flask/Fast API/Django backends, defining standards and contributing to strategic technical roadmaps.
- Built cloud-native full-stack applications on AWS Lambda, ECS, and Code Pipeline, applying TDD with Jest, Mocha, and Jasmine to maintain test coverage greater than 90%; monitored performance with CloudWatch and X-Ray.
- Implemented GraphQL and REST endpoints for seamless data communication; designed Redux Saga and MobX state management flows for complex async enterprise workflows.

UI Developer / React JS Developer | Alliance Insurance, Arlington, TX | Jan 2016 – Apr 2017

- Built SPAs using React, Redux, and Router with JWT auth and CSRF token security; created RESTful APIs with Node.js and Flask for data exchange, managed via Git, Bitbucket, and Jira in Agile workflows.

UI Developer | Commerce Benefit Group, Chennai, India | Jul 2013 – Feb 2014

- Built interactive UIs using Oracle APEX and web toolkits; developed complex PL/SQL models with cursors, performed Oracle database query and performance tuning, and scheduled backend jobs across enterprise data systems.
- Build mobile applications using iOS and Android.

UI Developer / Oracle PL/SQL Developer | Mepline Technical Services LLC, Dubai | Nov 2005 – Aug 2012

- Built interactive UIs using Oracle APEX and web toolkits; developed complex PL/SQL models with cursors, performed Oracle database query and performance tuning, and scheduled backend jobs across enterprise data systems.

EDUCATION

- MS in Information Technology, University of Cumberland, Kentucky, USA (2014–2015)
- MCA (Computer Science), Madurai Kamaraj University, India (2001–2004)
- Web Developer associate degree, MATC, Madison, WI
- BSc Physics, M S University, Tirunelveli, India (1998–2001)

CERTIFICATIONS

- Building smarter software: GenAI Tools and Practice- Coursera (GenAI for Front-End web developers, GenAI for Back-end developers, GenAI for mobile app developers – iOS, Android, ChatGPT API for



Developers: Integrate AI Effortlessly, GenAI for DevOps Practitioners, Introduction to GenAI for DevOps Practitioners, GenAI for Application Developers, Software creation: Version control, debugging, object design, Test management in software testing, GitLab AI: Mastering AI pair programming with GitLab Duo}

- AWS Generative AI and AI Agents with Amazon Bedrock
- Amazon Bedrock Customization, Optimization & Automation
- Getting Started with AWS Generative AI for Developers
- Generative AI Leader – Google Cloud
- AI Agent, AI Agents and context Eng, Tools in AI Agents – O'Reily
- AWS AI & ML Scholars - Udacity
- Meta full stack developer: Front-end and Back-end from scratch.
- Google UX design
- OCI – Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- OCA – Oracle PL/SQL Developer Certified Associate
- OCP – Oracle 11g Certified Professional, Advanced PL/SQL Developer
- OCA – Oracle 11g Certified Associate, Database Administrator
- Database and Big Data Analytics – Professional Certification, UC Santa Cruz
- Oracle Guided Learning Project Management Certified Foundations Associate
- Oracle Playbook for AI Excellence: Training and Assessment

Personal Contributions

- Portfolio
<https://hepzibha-portfolio.vercel.app/>
- Figma UI/UX design
<https://social-prize-54805720.figma.site/>
- GitHub projects
<https://github.com/Hepzibha2026>
- Tableau Dashboards
<https://public.tableau.com/app/profile/hepsi.ss/vizzes>
- AWS- Bedrock – Party Rock – Projects
<https://partyrock.aws/u/Hepzibha>